



2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = (r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(t) &= \int_{x_0}^x \begin{vmatrix} e^t & e^{2t} \\ e^{2t} & e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} dt \\ &= \int_{x_0}^x \frac{e^{t+2t} - e^{2t+t}}{e^{3t} + e^{2t}} dt \\ &= e^{2x} \int_{x_0}^x \frac{1}{e^{2t} + e^t} dt - e^x \int_{x_0}^x \frac{1}{e^t + 1} dt \end{aligned}$$

Meanwhile, since $\int \frac{1}{e^t + 1} dt = \int \frac{e^{-t}}{1 + e^{-t}} dt = -\log(1 + e^{-x})$

and $\int \frac{1}{e^{2t} + e^t} dt = \int \frac{e^{-t}}{e^t + 1} dt = \int \frac{-u}{\frac{1}{u} + 1} \cdot \frac{1}{u} du = \int \frac{-u}{1+u} du$

$$= \int -1 + \frac{1}{1+u} du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$\begin{aligned} y_p + C_1 e_1 + C_2 e_2 &= \left[e^x + e^{2x} \log(1+e^{-x}) + e^x \cdot \log(1+e^x) \right] \\ &+ C_1 e^x + C_2 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R}, \end{aligned}$$

2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p -subgroups.

By the Sylow theorem, $n_{101} \equiv 1 \pmod{101}$ and $n_{101} \mid 20$, so n_{101} must be 1. Therefore, the Sylow 101-subgroup is normal so proved.

Without using Sylow theorem.

By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.

Denote $A = \{g \mid g^{-1} \notin G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \rightarrow \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\#A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \rightarrow A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H -orbit of this action is $\text{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\#\text{Orb}_H(K) = |H : \text{Stab}_H(K)|, \text{Stab}_H(K) \leq H \Rightarrow |\text{Stab}_H(K)| \in \{1, 101\}.$$

So, $\#\text{Orb}_H(K) \in \{1, 101\}$ and $\text{Orb}_H(K) \leq A$, so $\#\text{Orb}_H(K) = 1$.

Therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.

Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{1\}$. That is, $H \leq G$ $\square$

2021. Spring

7. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \dots + a_d z^d \quad \text{over } \mathbb{C}, \quad a_d \neq 0$$

8m.

Evaluate the value $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$

Proof. By the fundamental theorem of Algebra, f has d -roots $z_1, \dots, z_d \in \mathbb{C}$.

That is, $f(z) = a_d(z - z_1) \dots (z - z_d)$

Moreover, $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \dots + \frac{1}{z - z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \dots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz = \sum_{n=1}^d \operatorname{Res}\left(\frac{z}{z - z_n}, z_n\right) = \sum_{n=1}^d \lim_{z \rightarrow z_n} z = z_1 + \dots + z_d$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \dots + z_d) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d}\right),$$

□

2022 Spring.

8. Define $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then, f is differentiable on $\mathbb{R}$.

Proof. For each $m \in \mathbb{N}$, define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first, $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$, because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence $f_m(x)$ converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if $|x| \leq 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if $|x| > 1$, then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives f_m converges uniformly.

Now, f_m converges at some point and f'_m converges uniformly on $\mathbb{R}$, the limit function f is differentiable. $\square$ sm.

First, for any $x \in \mathbb{R}$, $|x^2 + n^2| \geq n^2$, so

$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$

as $m \rightarrow \infty$. Therefore, given ε_0 , for some $N \in \mathbb{N}$, $m > n \geq N$ implies

$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$

Now, define for each $m \in \mathbb{N}$,

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

We already shown $f_m \rightarrow f$ uniformly. And,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

2022 Fall.

8. For continuous functions $g: [0,1] \rightarrow \mathbb{R}$ and $h: [-1,1] \rightarrow \mathbb{R}$, define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each $n \in \mathbb{N}$, define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that $f_n \rightarrow f$ uniformly on $[0,1]$

Proof. Define $l: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$.

Since h and g both are continuous, therefore l is continuous and since $[0,1]^2$ is compact set, so l is continuous uniformly.

Given $\epsilon > 0$, there exists $\delta > 0$ such that for any $(x_1, y_1), (x_2, y_2) \in [0,1]^2$,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|l(x_1, y_1) - l(x_2, y_2)\| < \epsilon.$$

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$, ($n > \frac{1}{\delta}$ gives $\frac{1}{n} < \delta$)

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 l(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} l(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} l(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} |l(x, y) - l(x, \frac{i}{n})| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \epsilon dy = \epsilon. \end{aligned}$$

Therefore, $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \epsilon$, So, proved. $\square$

12022. Fall.

18. Let S be the subset of $\mathbb{R}^3$ given by $\{(x, y, z) \mid z^2 = x^2 + y^2, z \geq 0\}$.

Show that S is not regular surface.

Proof. Suppose that S is a regular surface. Then, for $(0, 0, 0) \in S$, there exists a neighbor V containing $(0, 0, 0)$ such that $V \cap S$ is a graph of

$$z = f(x, y) \quad \text{or} \quad y = g(x, z) \quad \text{or} \quad z = h(y, z)$$

where f, g, h are differentiable. Since projection of S into xz , yz -plane are not 1-1, g, h cannot be such a map.

Meanwhile, $f(x, y) = \sqrt{x^2 + y^2}$ at locally $(0, 0, 0)$, but f is not differentiable at $(x, y) = (0, 0)$, so it cannot be possible. Therefore S is not regular.

(More specifically, $(0, \pm \epsilon, \sqrt{1 - \epsilon^2}) \in S$ for any $\epsilon > 0$, so there is no 1-1 yz -projection, similarly xz .)

2023, Spring.

7. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be an entire satisfying 10m

$$U^2(z) - 2 \cdot V^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. Observe that ~~differentiating gives~~ that

$$2U U_x - 2V V_x = 0 \Rightarrow \cancel{U U_x} = V V_x = -V \cdot U_y$$

Observe that: $f(z) = 0 \Leftrightarrow U(z) = V(z) = 0$

But, $U^2 - 2V^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f(z)} = \frac{1}{U^2(z) + V^2(z)} = \frac{1}{1 + 3V^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, so f is also constant function. $\square$

$$\text{Or, } |f^2| = U^2 + V^2 = 1 + 3V^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3V^2}} \leq 1.$$

2024, Spring.

1. Let $A, B \in M_{m \times n}(F)$. Show that $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$. 5m

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given $\alpha \in \text{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$ for some $\beta \in F^n$, therefore $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$. Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ such that $P\alpha_i = C_i\alpha_i$ for some $C_i > 0$, for each i .

Let $1 \leq i \leq n$ be fixed. Then, by the assumption,

5m

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - C_i I)\alpha_i = 0$, because: if it is not zero, then $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$,

so $-C_i$ is negative eigenvalue of Q , it contradicts to Q is positive definite. Therefore, for each $1 \leq i \leq n$,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. $\square$

2024, Spring

3. Let $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T -invariant subspace $V \subseteq \mathbb{R}^5$ s.t. $\dim V = 4$?

b). "

s.t. $\dim V = 2$, $T|_V$ is diagonalizable?

c). "

$\dim V = 3$, "

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in V , so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of T is $(t-1)^5$.

If such a subspace V exists, then the characteristic polynomial of $T|_V$ is

$(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.

That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

1/22/24 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$. For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f, g) := \sup_{x \in \mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n \rightarrow \infty} d(f, f_n) = 0$

b). Prove that $\lim_{n \rightarrow \infty} \left(\sup_{x \in \mathbb{R}} |f_n(x)| \right) = \sup_{x \in \mathbb{R}} |f(x)|$.

[Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis. Since $\mathbb{R}$ is complete, $\lim_{n \rightarrow \infty} f_n(x)$ exists, put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \rightarrow 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x \in \mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$.

So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \rightarrow \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$.

Since x was arbitrary, $\sup_{x \in \mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.

Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$

and since f_n was bounded, f is bounded. And, the argument above

proves $\sup_{x \in \mathbb{R}} |f - f_n| \rightarrow 0$ as $n \rightarrow \infty$, so since f_n is continuous and

$f_n \rightarrow f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left| \sup_{x \in \mathbb{R}} |f_n(x)| - \sup_{x \in \mathbb{R}} |f(x)| \right| \leq \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$

Prove directly.

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

8m

b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

2025. Fall

2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

15m: A^n 의 rank, $n=2$ 일 때부터 시작

a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

2025. Fall

3. Find all $a, b, c \in \mathbb{R}$ such that

6m

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

2025, Fall,

4, Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

20m

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol), Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \mathcal{Q}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 구조 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

12-25. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b). By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g , $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \pmod{n}$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solutions.

2025. Fall.

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0, 1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.

Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.

Let α be a limit point of $\{a_n\}^*$. Then, given $\epsilon \in \mathbb{R}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

For $k=1$, choose such a $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{\frac{1}{2}}(p)$, then $a_{p_1} \in B_1(\alpha)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

Then, similarly, there is $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$ such that $p_k > p_{k-1}$.

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point α is contained in $\{a_n\}^*$.

so it is closed.

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < \alpha + \epsilon$ (since $\alpha = \max \{a_n\}^*$) and there exists infinitely many terms in $\{a_n\}$ such that $\alpha - \epsilon < a_n$ (since $a_{p_k} \rightarrow \alpha$)

Thus $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0, 1]$, there exists a subsequence in $\{b_n\}$ converges to b , because: for any $\epsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let E be a nonempty bounded subset of a Metric space (X, d) .

10m

Define $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove $\text{diam } E = \text{diam } \bar{E}$

b). If E is compact, then $d(x, y) = \text{diam } E$ for some $x, y \in E$.

Proof.

a). Since $E \subseteq \bar{E}$, $\text{diam } E \leq \text{diam } \bar{E}$ is clear.

To show the inverse inequality, using the property of supremum.

Given $\varepsilon > 0$, there exists $x, y \in \bar{E}$ such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since $\bar{E}$ is closure, there are $x', y' \in E$ such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now, $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\text{diam } \bar{E} \leq \text{diam } E$, so proved. 7m.

b). Since E is compact set, so $E \times E$ is also compact set.

Since the metric function d is continuous, it preserves compactness, so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is $(x', y') \in E \times E$ such that $d(x', y') = \max d[E \times E]$ 3m.

2025 Fall.

8. Let $a > 1$. Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now,
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because $\sum (-1)^n x^{2n-1}/(2n)!$ converges for all $x \in \mathbb{R}$)

Now,
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0, \text{ so}$$

the given limit is $\ln a$.

Revised in next

Observation for $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$.

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all $x \neq 0$. Therefore, for any $t > 0$,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \Big|_t^a \end{aligned}$$

Meanwhile, the series converges uniformly in $[t, a]$ for small enough $t > 0$, specifically, $t < \frac{1}{a}$, then for each $n \in \mathbb{N}$,

$$\left| (-1)^n \frac{x^{2n}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{2n \cdot (2n)!} \cdot (a^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note: $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$, $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} \rightarrow 0$ as $t \rightarrow 0$.

Fall.

12. Define an equi. rel on $\mathbb{R}$ as $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$.

" as $x \sim y \Leftrightarrow x = y$ or $x - y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\sim$.

c). In $\mathbb{R}/\sim$, determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since $\mathbb{Q}$ is closed under the operation $-$, so these relations are equi. rel.

a). First, $\mathbb{R}/\sim = \{a + \mathbb{Q} \mid a \in \mathbb{R}\}$, where $a + \mathbb{Q} = \{a + q \mid q \in \mathbb{Q}\}$. And,

$U \subseteq \mathbb{R}/\sim$ is open $\Leftrightarrow \pi^{-1}[U]$ is open, where $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim : a \mapsto a + \mathbb{Q}$.

ΓKLG , 73p.

2.7.14.

If $\{x_n\} \subset B$ converges to $x \in B$, then

$$\bigcap_{k=1}^{\infty} \{x_n \mid n \geq k\} = \{x\}.$$

Γ_{KKG} ,

2.7.7, and 2.7.20.

Let $A, B \subseteq \mathbb{R}^n$.

1) If A is open, then $A+B$ is open.

2) If A, B

TSNU. Q. 14.

Prove that $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$ exists.

Proof. First, let $a_n = n\pi$ ($n \in \mathbb{N}$). Then, a sequence $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$ converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$ is alternating series, so converges.

7226. Prima.

3. Let V be an n -dimensional vector space over $\mathbb{C}$,
Let $T: V \rightarrow V$ be a linear operator such that $T^3 = T$.

27m

a). Find all possible characteristic polynomials and corresponding minimal polynomials for T .

b). Prove $V = \ker T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.

$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$$m(t) = t \text{ or } t-1 \text{ or } t+1 \text{ or } t(t-1) \text{ or } t(t+1) \text{ or } (t-1)(t+1) \text{ or } t(t-1)(t+1).$$

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \text{ where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is $(t-1)(t+1)$

and other cases are similar.

b). ~~Given $\alpha \in \ker T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$,
so $\alpha - T^2\alpha \in \ker T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover, if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$,
Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = TT^2\beta = \alpha = 0$,
so $V = \ker T \oplus \operatorname{Im} T$.

Modified.

Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.

a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Describe $T|_{\operatorname{Im} T^2}$.

c). Prove that T is diagonalizable $\Leftrightarrow \ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

so $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. so, $T\alpha = T^3\beta$.

Multiplying T , then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$.

Therefore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$, $\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$, so, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$, moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable. That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$.

2026 Fall.

4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T: G \rightarrow S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\text{Ker } T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

$$\begin{aligned} \text{a). Given } g_1, g_2 \in G, T(g_1 g_2)(aH_1, bH_2) &= (g_1 g_2 aH_1, g_1 g_2 bH_2) \\ &= T(g_1) \circ T(g_2)(aH_1, bH_2) \end{aligned}$$

for each $a, b \in G$, so it preserves the operation.

$$\text{And } g \in \text{Ker } T \Leftrightarrow T_g = \text{id} \Leftrightarrow \text{for all } a, b \in G, T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$$

$$\begin{aligned} &\Leftrightarrow \text{for all } a, b \in G, a^{-1}ga \in H_1 \text{ and } b^{-1}gb \in H_2 \\ &\Rightarrow g \in H_1 \cap H_2 \text{ (take } a=b=1_G) \end{aligned}$$

b). By the first isomorphism theorem,

$$G/\text{Ker } T \cong \text{Im } T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

$$\text{Now, } |G : H_1 \cap H_2| \leq |G : \text{Ker } T| \leq \# S_{G/H_1 \times G/H_2}$$

because $\text{Ker } T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G : A|$ is finite, then

$$G/A \rightarrow G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \leq B$,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

12.26. Fall.

8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each $n \geq 1$, define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$, $x \in \mathbb{R}$.

a). Show that $T_n f \rightarrow f$ on $[0,1]$.

b). Determine $T_n f \rightarrow f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n}+x \leq y \leq \frac{1}{n}+x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore, $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$ as $n \rightarrow \infty$, so converges uniformly.

